What we added to process four

Supplied slides

Shortlisting-rate gaps
Activation cosine and norm metrics

Our implemented additions

Matched names and fixed-pool top-k tests
Complete head states and paired distances
Disjoint correction fit, probe train and test
Usefulness + coverage gates with verdicts

A runnable audit for Qwen 1.5B and Clef 9B

Original upstream cues remain. Whole-model erasure stays unproved.

The correction and evaluation process

The fixed-family correction

Fixed CV and job, four naming cues
Capture complete decision inputs
$h^* = \Sigma_a \pi_a h_a$
Original head receives the same h^*

Qwen: 1,536-vector for A/B readout
Clef: complete T × 4,096 state + fixed head metadata

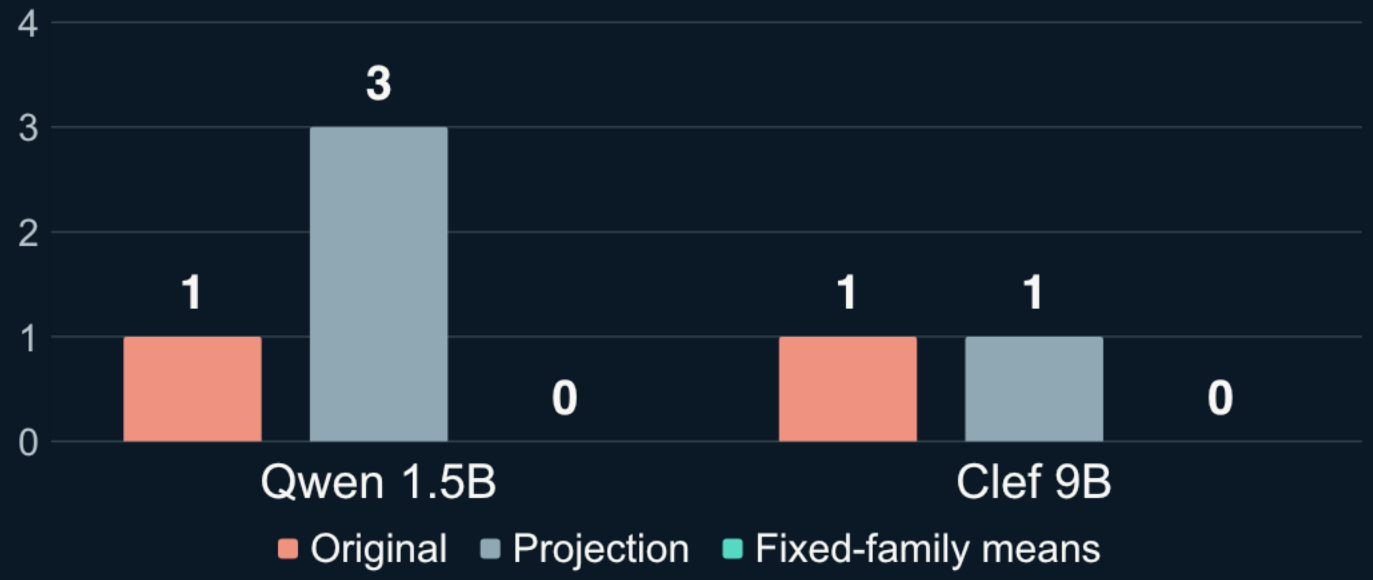
Independent audits

Paired scores, ranks and top-k
Activation distances + probes
Usefulness and coverage
Pass / fail / insufficient evidence

No: behavioral invariance alone does not imply activation erasure

A correction can pass one test and fail another

Pools where names changed the selected applicant



4-bit LoRA failed

Presence: 0.0417 to 0.0190

Unstable pools: 2/6 to 3/6
Cue-probe accuracy: 100%

6 convenience pools, 16 held-out CV excerpts, top 1. Hiring accuracy unmeasured.

Answers to the challenge brief

How do we ensure the model is truly unbiased?

Scoped name invariance at the complete interface and fixed-pool ranking.
Independent output, activation and utility gates. Global fairness remains unproved.

What are the sociotechnical risks?

Qualification proxies and unknown usefulness can hide unfair selection.
Safeguards: reviewed rubrics, privacy, human oversight and accessible appeals.

How did we use the datasets?

Kaggle CVs and LinkedIn criteria stay fixed. Verified race / hiring-quality labels unavailable.